Kidra CPU reference: multi-window transformer with shared global exchange

Architecture kidra-reference/2 • one synchronized decision snapshot per tick

(a) Model overview

Open windows $w \in \mathcal{W}_t$
Local positions p_w; slots $s_w < S$

Memory: $E_{\text{tok}}[z] + b_{\text{id}} E_{\text{win}}[s_w]$
Query: $e_q + b_{\text{id}} E_{\text{win}}[s_w]$

L local blocks + global exchanges
Row states $h^{(\ell)} \in \mathbb{R}^d$
Memory and queries obey the same masks

RMSNorm $\rightarrow W_{\text{vocab}}$
Query logits $\in \mathbb{R}^V$

Unshifted teacher-forced targets

$$\mathcal{L} = \frac{\sum_r \omega_r \, \text{CE}(a_r, z_r)}{\max(\sum_r \omega_r, 1)}$$

Prediction queries never embed their targets.
Only durable tokens populate memory.

(b) Expanded layer $\ell = 1, \ldots, L$

Input row state $h^{(\ell-1)} \in \mathbb{R}^d$

Local attention + residual
$Q = \text{RoPE}_{p_w}(W_Q^\ell \text{RMSNorm}(h))$
$u = h + W_O^\ell \text{Attn}(Q, K_w, V_w; M_w)$

SwiGLU feed-forward + residual
$f = u + W_\downarrow^\ell \, [\text{SiLU}(W_g^\ell \bar{u}) \odot W_\uparrow^\ell \, \bar{u}]$
$\bar{u} = \text{RMSNorm}(u), \quad d_{\text{ff}} = 2d$

Shared global exchange + residual
$Q_g = W_{Q,g} \text{LayerNorm}(f)$
$h^{(\ell)} = f + W_{O,g} \text{Attn}(Q_g, K_g, V_g; M_g)$
No global K/V projections or query RoPE

Output $h^{(\ell)}$ to layer $\ell + 1$ (or head)

Durable memory layer inputs
$\{h_j^{(\ell-1)}, p_j, s_j\}$

Reuse local projections
$K_j = \text{RoPE}_{p_j}(W_K^\ell \bar{h}_j)$
$V_j = W_V^\ell \bar{h}_j$

Global memory bank
$K_g = \text{cat}_j(K_j + b_{\text{id}} E_{\text{win}}[s_j])$
$V_g = \text{cat}_j(V_j)$

Own + other visible windows
Local RoPE is retained on K.
Identity is added only to K.

$$\text{Attn}(Q, K, V; M) = \text{softmax}(QK^T/\sqrt{d_h} + M)V$$

(c) Tick and memory semantics

A Open windows
Insert BOS and headers

B Ordered observations
Predict supervised B before insertion

C Shared after-B snapshot
Compute all D logits and memory states

D Atomic commit
Durable targets enter memory; NOOP does not

Evict / advance
Release closed / replaced cache entries

No current-tick D target is a memory key.
Every layer enforces causal visibility.

Solid arrows: state flow. Dashed arrows: memory reads. Exchange weights are shared across layers; local-block weights are layer-specific.